ACADEMIC CURRICULUM VITAE

Devrim Tümer

Ph.D. Candidate in Mechanical Engineering

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Mechanics. Materials. Computation.

Research in computational biomechanics and constitutive modelling of soft biological and biomimetic materials. Interests include hyperelasticity, viscoelasticity, finite element analysis, parameter identification and microstructure-informed modelling of fibre-reinforced materials.

01 Research appointments

2025 and 2026
Research visits

Guest Researcher

Institute of Biomechanics, Graz University of Technology, Austria

August - October 2025; February - December 2026

March 2020
to present

Research and Teaching Assistant

Department of Mechanical Engineering, Ege University, Türkiye

02 Education

2021 - present

Ph.D. in Mechanical Engineering · ongoing

Ege University, Türkiye

2018 - 2021

M.Sc. in Mechanical Engineering

Dokuz Eylül University, Türkiye

Thesis: Analyzing and assessment of different muscle groups in shoulder joint with finite element method.

2012 - 2017

B.Sc. in Mechanical Engineering

Dokuz Eylül University, Türkiye

03 Methods and scientific computing

Mechanics: constitutive modelling; finite element methods; hyperelasticity and viscoelasticity; multiscale and representative volume element modelling; parameter identification.
Software and programming: Abaqus, FEBio, Python, MATLAB and C.

Publications & academic activity

04 Selected peer-reviewed publications

2026 **Yumuşak Dokunun Biyomekanik Karakterizasyonu için iki Eksenli Çekme-Basma Test Cihazı**

The Biaxial Tension-Compression Test Device for the Biomechanical Characterization of Soft Tissue

Schirinova, S., Sapmaz, S., **Tümer, D.**, Harmanşah, C., & Pekedis, M.

Dokuz Eylül Üniversitesi Mühendislik Fakültesi Fen ve Mühendislik Dergisi, 28(82), 77-82.

doi.org/10.21205/deufmd.2026288210

2024 **The effects of tears in infraspinatus on other rotator cuff constituents**

Tümer, D., Arman, Y., & Havıtçıoğlu, H.

Journal of the Mechanical Behavior of Biomedical Materials, 157, 106600.

doi.org/10.1016/j.jmbbm.2024.106600

2024 **Design and heat analysis of basalt glass as a thermal storage material in a concentrated solar power (CSP) system**

Günindi, B., Toygar, M. E., & **Tümer, D.**

Journal of Thermal Analysis and Calorimetry, 149(5), 2007-2023.

doi.org/10.1007/s10973-023-12815-9

2020 **Investigation of effective coating of the Ti-6Al-4V alloy and 316L stainless steel with graphene or carbon nanotubes with finite element methods**

Tümer, D., Güngörürler, M., Havıtçıoğlu, H., & Arman, Y.

Journal of Materials Research and Technology, 9(6), 15880-15893.

doi.org/10.1016/j.jmrt.2020.11.052

05 Teaching and academic service

Teaching assistance · Ege University
Statics; Strength of Materials; Computational Mechanics; Engineering Materials; Machine Laboratory.

Selected reviewing and professional development
Engineering Research Express (2024); Journal of Materials Research and Technology (2023). IOP Trusted Reviewer certificate (2024).